

01

Cloud Waste from Forgotten VMs

- Dev/test VMs get spun up for a task and forgotten — billing continues even when idle
- Standard cost dashboards report waste after the money is already gone — reactive, not preventive
- A student/small-team account has no dedicated FinOps person watching this

8 hrs/day

Typical dev VM usage
(9am – 5pm)

24 hrs/day

Typical billing
(if left running)

02

Smart Cloud Budget Guard

A preventive assistant — not another after-the-fact cost report

1

Monitor

Inventory Azure VMs and persist their state

2

Detect Idle

Flag low-CPU / low-network VMs via Azure Monitor

3

Track Budget

Define monthly budgets and thresholds

4

Alert

Raise alerts on idle or budget breaches

5

Automate

Safely recommend / stop tagged VMs

PLANNED

Finalized Tech Stack

Layer	Technology
Backend framework	FastAPI (Python 3.11+)
Azure SDK	azure-identity, azure-mgmt-compute, azure-mgmt-monitor
Authentication	DefaultAzureCredential via Azure CLI login
Database	PostgreSQL, via SQLAlchemy (sync) + psycopg (v3)
Schema management	Plain SQL in docs/db_schema.sql
Scheduler	APScheduler (in-process BackgroundScheduler)
Frontend	React + Vite, Recharts (planned)

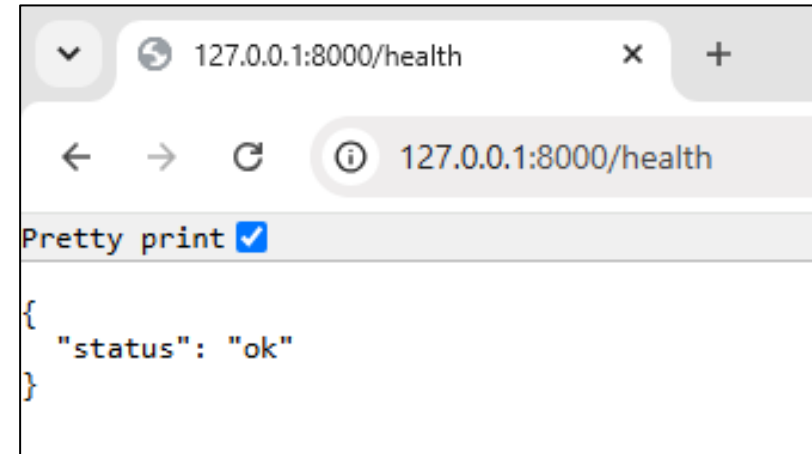
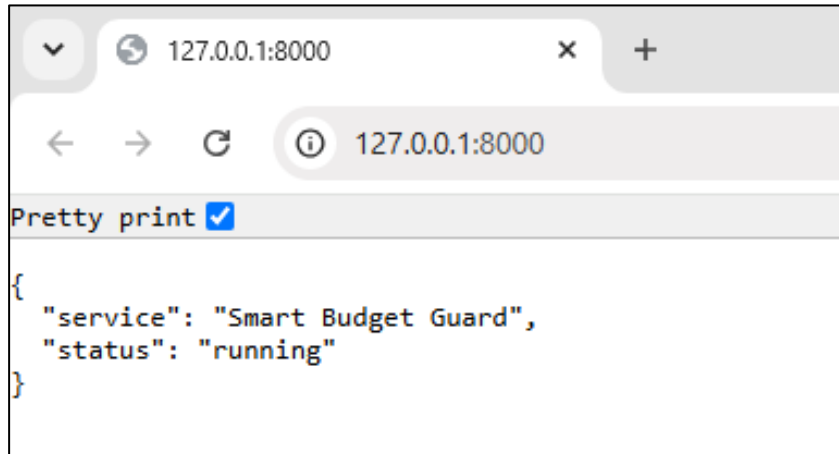
Project Structure

- Modular backend package: routers, services, and database layers separated
- backend/app/ — main.py, vms.py, budgets.py, database/, services/
- scripts/ — standalone Azure VM listing script
- docs/ — schema definitions and design notes

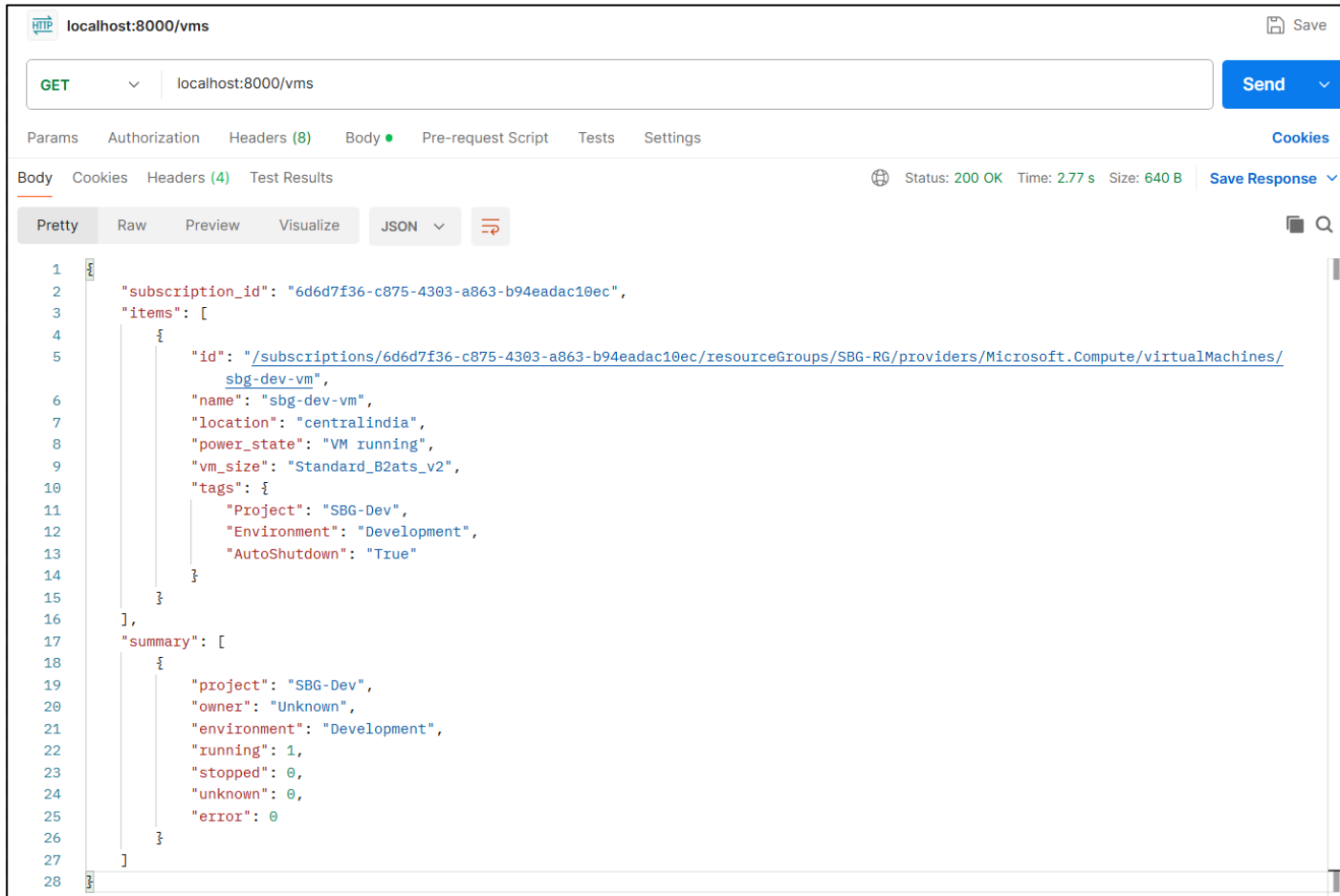
```
▼ smart-cloud-budget-guard
  ▼ backend
    > __pycache__
    ▼ app
      > __pycache__
      > database
      > services
      + __init__.py
      + budgets.py
      + main.py
      + vms.py
      + __init__.py
    ▼ docs
      + db_schema.sql
    ▼ scripts
      > __pycache__
      + __init__.py
      + list_vms.py
    > venv
    + .env
    + .gitignore
    + Documentation.md
    + README.md
    + requirements.txt
```

Backend Running + Health Checks

```
(venv) PS C:\Users\pkv03\College\Semester 3\Minor Project\smart-cloud-budget-guard> uvicorn backend.app.main:app --reload
INFO: Will watch for changes in these directories: ['C:\\Users\\pkv03\\College\\Semester 3\\Minor Project\\smart-cloud-budget-guard']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [17428] using StatReload
INFO: Started server process [25916]
INFO: Waiting for application startup.
Starting scheduler...
INFO:apscheduler.scheduler:Adding job tentatively -- it will be properly scheduled when the scheduler starts
INFO:apscheduler.scheduler:Added job "collect_and_store_metrics" to job store "default"
INFO:apscheduler.scheduler:Scheduler started
DEBUG:apscheduler.scheduler:Looking for jobs to run
Scheduler Jobs: [<Job (id=metrics_collection name=collect_and_store_metrics)>]
INFO:apscheduler.executors.default:Running job "collect_and_store_metrics (trigger: interval[1:00:00], next run at: 2026-09-06 22:06:53 IST)" (scheduled at 2026-09-06 22:06:53.329926+05:30)
INFO: Application startup complete.
DEBUG:apscheduler.scheduler:Next wakeup is due at 2026-09-06 23:06:53.329926+05:30 (in 3599.997372 seconds)
INFO:apscheduler.executors.default:Job "collect and store metrics (trigger: interval[1:00:00], next run at: 2026-09-06 23:06:53 IST)" executed successfully
```



Live Azure VM Inventory



The screenshot shows a REST client interface with the following details:

- URL:** localhost:8000/vms
- Method:** GET
- Status:** 200 OK
- Time:** 2.77 s
- Size:** 640 B

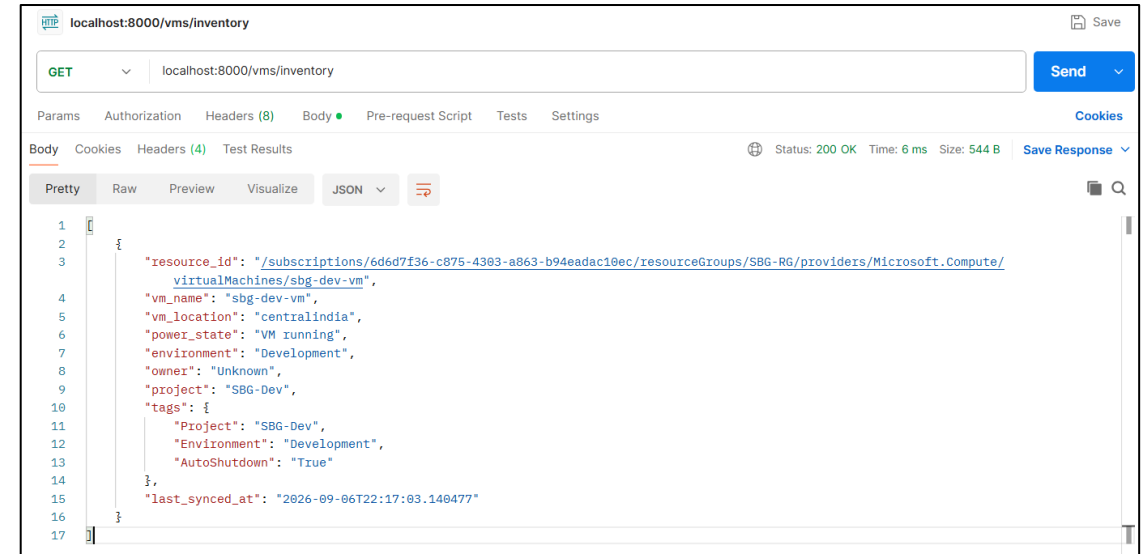
The response body is displayed in JSON format:

```
1 {
2   "subscription_id": "6d6d7f36-c875-4303-a863-b94eadac10ec",
3   "items": [
4     {
5       "id": "/subscriptions/6d6d7f36-c875-4303-a863-b94eadac10ec/resourceGroups/SBG-RG/providers/Microsoft.Compute/virtualMachines/sbg-dev-vm",
6       "name": "sbg-dev-vm",
7       "location": "centralindia",
8       "power_state": "VM running",
9       "vm_size": "Standard_B2ats_v2",
10      "tags": {
11        "Project": "SBG-Dev",
12        "Environment": "Development",
13        "AutoShutdown": "True"
14      }
15    }
16  ],
17  "summary": [
18    {
19      "project": "SBG-Dev",
20      "owner": "Unknown",
21      "environment": "Development",
22      "running": 1,
23      "stopped": 0,
24      "unknown": 0,
25      "error": 0
26    }
27  ]
28 }
```

- Authenticated via Azure CLI login (DefaultAzureCredential)
- No client secrets stored anywhere — required on a student subscription
- Returns real-time power state and tags per VM

Persistence Layer (PostgreSQL)

```
postgres=# \c smart_budget_guard
You are now connected to database "smart_budget_guard" as user "postgres".
smart_budget_guard=# \dt
          List of tables
 Schema |   Name   | Type | Owner
-----+-----+-----+-----
 public | budgets  | table | postgres
 public | resources | table | postgres
 public | vm_metrics | table | postgres
(3 rows)
```



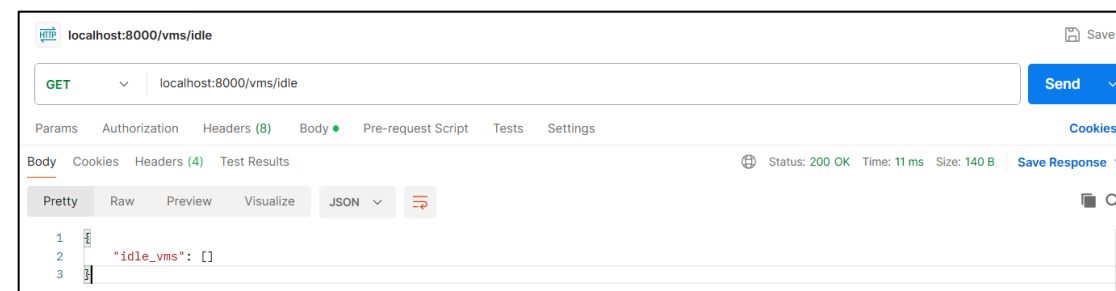
- Live Azure data is synced and upserted into PostgreSQL — inventory no longer depends on a live Azure call every request

Metrics Collection + Idle Detection

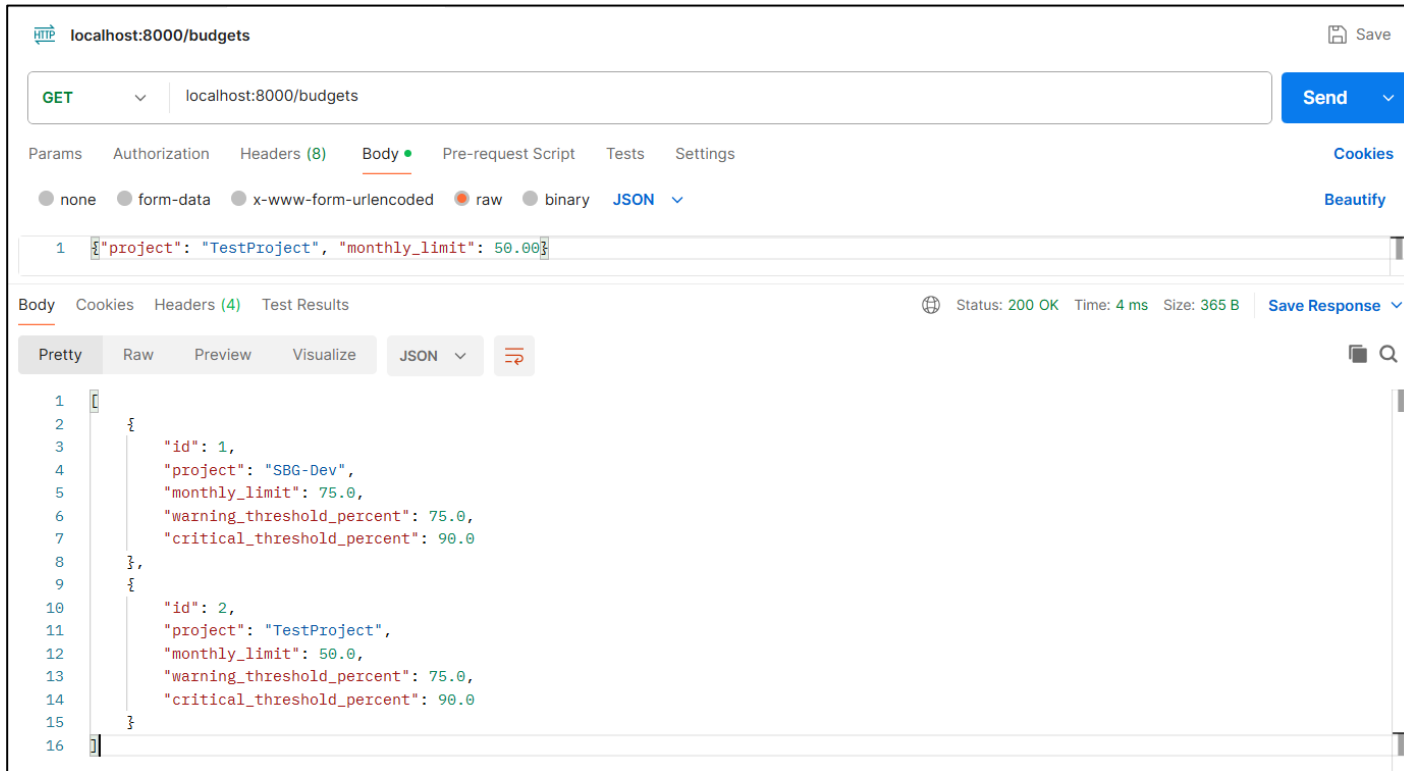
```
smart_budget_guard=# select * from vm_metrics;
```

id	vm_name	cpu_percent	network_in_bytes	network_out_bytes	recorded_at
1	sbg-dev-vm	0.16	2553610	3824943	2026-08-13 14:17:11.414831
4	sbg-dev-vm	0.15	2510111	3859980	2026-08-15 16:08:58.367438
5	sbg-dev-vm	0.15	2510025	3859926	2026-08-15 16:11:59.099509
6	sbg-dev-vm	0.15	2510025	3859926	2026-08-15 16:12:59.05911
7	sbg-dev-vm	0.15	2510025	3859926	2026-08-15 16:13:59.072327
8	sbg-dev-vm	0.15	2510025	3859926	2026-08-15 16:14:59.069657
9	sbg-dev-vm	0.15	2510025	3859926	2026-08-15 16:15:59.071587
10	sbg-dev-vm	0.15	2510025	3859926	2026-08-15 16:16:59.060385
11	sbg-dev-vm	0.15	2513955	3868341	2026-08-15 17:09:55.959232
12	sbg-dev-vm	0.15	2513955	3868341	2026-08-15 17:10:55.933689
13	sbg-dev-vm	0.15	2516163	3871964	2026-08-15 18:54:23.201784
14	sbg-dev-vm	0.15	2515733	3872245	2026-08-15 19:03:08.633336
15	sbg-dev-vm	0.16	2519429	3877584	2026-08-15 19:10:27.584736
16	sbg-dev-vm	0.15	2529890	4172686	2026-09-04 15:54:04.687309
17	sbg-dev-vm	0.15	2524982	4164992	2026-09-04 16:27:01.352684
18	sbg-dev-vm	0.15	2524916	4164992	2026-09-04 16:29:18.207149
19	sbg-dev-vm	0.15	2524916	4164992	2026-09-04 16:30:01.384434
20	sbg-dev-vm	0.15	2525048	4164992	2026-09-04 16:31:00.652115
21	sbg-dev-vm	0.15	2525048	4164992	2026-09-04 16:32:41.018083
22	sbg-dev-vm	0.15	2531255	4178517	2026-09-04 16:58:51.760873
23	sbg-dev-vm	0.15	2531255	4178517	2026-09-04 16:58:57.519564

-- More --



- Background scheduler collects CPU + network metrics hourly — idle rule: CPU < 5% and network < 5 KB/s, sustained over 2 hours



The screenshot shows a REST client interface with the following details:

- URL:** localhost:8000/budgets
- Method:** GET
- Body:** [{"project": "TestProject", "monthly_limit": 50.00}]
- Status:** 200 OK
- Time:** 4 ms
- Size:** 365 B
- Response Body (JSON):**

```
[
  {
    "id": 1,
    "project": "SBG-Dev",
    "monthly_limit": 75.0,
    "warning_threshold_percent": 75.0,
    "critical_threshold_percent": 90.0
  },
  {
    "id": 2,
    "project": "TestProject",
    "monthly_limit": 50.0,
    "warning_threshold_percent": 75.0,
    "critical_threshold_percent": 90.0
  }
]
```

- Mock cost model: hourly rate by VM size × running hours this month
- Real Azure Cost Management access isn't reliable on a student subscription
- Budget state: normal / warning / critical, based on spend-to-limit ratio

Challenges Solved

1

psycopg2 DLL load failure on Python 3.14

No compatible prebuilt wheel for a very new interpreter → migrated to psycopg (v3), which has proper Python 3.14 support.

2

Naive 1 KB/s idle network threshold never matched

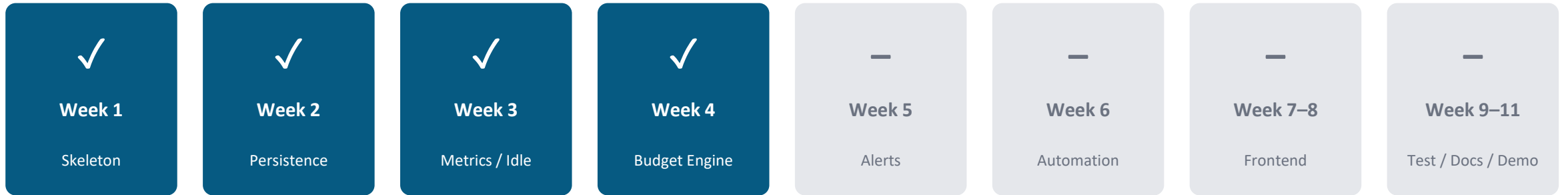
Measured a fully idle VM's real background traffic (~1.75 KB/s from Azure agent/heartbeat overhead) → tuned the threshold to 5 KB/s.

3

Budget spend silently returned \$0

Traced to a mismatch between the Azure VM's Project tag and the budget's project field → fixed at the source by standardizing the tag

Current State and Future Works



4 of 11 weeks complete

- Backend inventory, persistence, metrics, and budget tracking are all working end-to-end against real Azure resources
- Next milestone: alerts — turning idle/budget signals into something actionable